

HARSHITA SINGH

Jamshedpur, Jharkhand

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Education

Vellore Institute of Technology

2023 – Present

B.Tech in Computer Science and Engineering

CGPA: 9.00

Experience

FRONTEND DESIGNER INTERN:

May 2025 – June 2025

Centre for Disaster Mitigation and Management (CDMM)

Under Dr. Surendar M

- Completed an intensive summer internship focused on disaster management technologies and analytical problem solving.
- Collaborated with mentors to understand real-world applications of technology in disaster risk mitigation.
- Received Certificate of Appreciation for outstanding contribution and performance.

SOFTWARE FULL STACK INTERN

2025

Machine Vision and Intelligence Lab, NIT Jamshedpur

- Designed and implemented a high-precision architecture for rich text rendering in modern web editors.
- Developed a reusable rendering framework adaptable to multiple web applications.
- GitHub Repository: github.com/Harshita1803/NITPR

Projects

Smart Agricultural Supply Chain System | Full Stack Web Application

Smart India Hackathon Finalist

- Developed a full-stack logistics platform to manage agricultural product transportation efficiently.
- Designed modules for farmers, transporters, and buyers to streamline product movement.
- Improved supply chain workflow by reducing delays and increasing transparency.

Smart Campus Navigator | Web Application

DevJams (GDSC-VIT)

- Developed an interactive campus navigation platform to improve accessibility.
- Integrated location-based navigation and optimized route guidance.
- Live Demo: www.devjams.gdscvit.com

Safe Driving Monitoring System | Web + Sensors

Caterpillar Hackathon

- Developed a driver safety monitoring system capable of detecting unsafe driving behaviour in real time.
- Integrated sensor data to improve driving awareness and accident prevention.

Tic-Tac-Toe Game | React.js

Gravitas Finalist

- Built a responsive two-player Tic-Tac-Toe application using React.js.
- Implemented efficient game logic with real-time winner detection and restart functionality.

AI & NLP Automation Suite | Python, n8n, LLM, Neo4j

- Developed an NLP-powered workflow using n8n and Large Language Models to retrieve structured information from datasets using natural language queries.
- Built a domain-specific chatbot using Python, TF-IDF Vectorization, Cosine Similarity, JSON Corpus, and NLTK for intelligent question answering.
- Designed a Knowledge Graph using Neo4j and Cypher queries to perform graph-based reasoning and relationship discovery.
- Implemented corpus creation, text preprocessing, retrieval-based question answering, and workflow orchestration using modern NLP techniques.

Adaptive Checkpoint-Resume Framework for Stateful Execution in Serverless Computing | Node.js, Express.js, Cloud

- Designed a predictive runtime monitoring framework enabling checkpoint-resume execution for long-running serverless workloads.
- Implemented dynamic state serialization and recursive execution to overcome serverless timeout limitations.
- Developed a prototype using Node.js, Express.js and JSON-based state management with integrated API security validation.
- Focused on improving reliability, minimizing recomputation overhead and enabling pseudo-stateful execution in stateless serverless environments.

Achievements

- Finalist – Caterpillar Hackathon
- Finalist – Smart India Hackathon
- Finalist – Subway Cypher at Gravitas
- Shortlisted – DevJams'24 by GDSC VIT

Technical Skills

Languages	Java, JavaScript, Python, HTML, CSS
Frameworks	React.js, Node.js, Express.js, Next.js, Tailwind CSS
Databases	MySQL, JSON
Cloud & Tools	Git, GitHub, n8n, Neo4j, VS Code
Core Concepts	Data Structures and Algorithms, Object-Oriented Programming, Problem Solving, NLP, Machine Learning Basics, Cloud Computing, Knowledge Graphs

Leadership Activities

- Senior Core Member – VIT Anchoring Club
- Senior Core Member – VIT Geospatial Club

Interests

Coding, Algorithms, Artificial Intelligence, Cloud Computing, Problem Solving, Reading